

20.1 Introduction

$H \varphi_n(x) = \lambda_n \varphi_n(x)$; n integer $\Rightarrow$ discrete spectrum
 $\downarrow$
 Hermitian $\{ \varphi_n \}$ discr. orthonormal basis

$$\langle \varphi_n | \varphi_m \rangle = \delta_{nm}$$

Expansion: $g(x) = \sum_n f_n \varphi_n(x) \Rightarrow f_n = \int_a^b \varphi_n^*(x) g(x) dx$

$n \rightarrow$ cont. $t \Rightarrow$ continuous spectrum $\Rightarrow$ Dirac - delta orthonormality

$$\langle \varphi_t | \varphi_{t'} \rangle = \delta(t - t')$$

Integral repr.:

$$g(x) = \int_a^b f(t) \varphi(x, t) dt$$

$$f(t) = \int_a^b g(x) \varphi^*(x, t) dx$$

Integral transform

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$$g(x) = \int_a^b f(t) \underbrace{K(x, t)}_{\text{kernel}} dt$$

$$g(x) = \mathcal{L} f(t)$$

$$\mathcal{L}^{-1} g(x) = f(t)$$

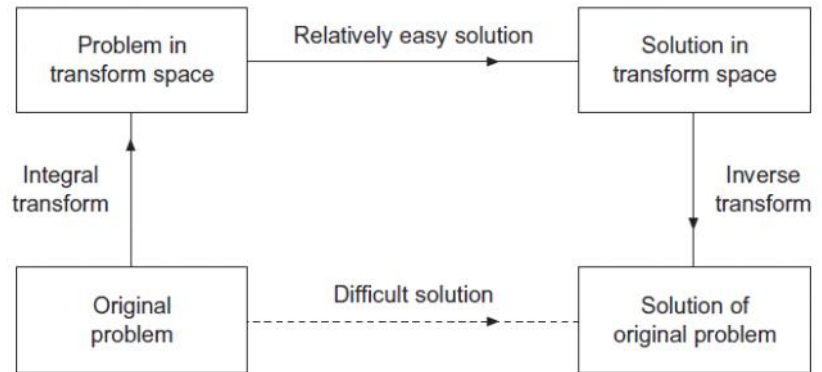


FIGURE 20.1 Schematic: use of integral transforms.

Some Integral Transformations

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Fourier transform

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt$$

Laplace transform

$$f(s) = \int_0^{\infty} e^{-ts} F(t) dt$$

Hankel transform

$$g(\alpha) = \int_0^{\infty} f(t) t J_n(\alpha t) dt$$

Mellin transform

$$g(\alpha) = \int_0^{\infty} f(t) t^{\alpha-1} dt$$

ex. $g(\alpha) = \Gamma(\alpha) \Leftrightarrow f(t) = e^{-t}$

20.2 Fourier transform

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$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{j\omega t} dt$$

Ex 20.2.1

$$f(t) = e^{-\alpha |t|}$$

$$\Rightarrow g(\omega) = \frac{1}{\sqrt{2\pi}} \frac{2\alpha}{\alpha^2 + \omega^2}$$

$$f(t) = \delta(t)$$

$$\Rightarrow g(\omega) = \frac{1}{\sqrt{2\pi}}$$

$$f(t) = 2\alpha \frac{1}{\sqrt{2\pi}} \frac{1}{\alpha^2 + t^2} \Rightarrow g(\omega) = \frac{e^{-\alpha |\omega|}}{e}$$

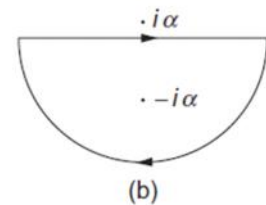
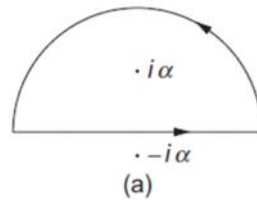


FIGURE 20.2 Contours for third transform in Example 20.2.1.

Ex 20.2.2

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$$f(t) = e^{-at^2}$$

$$g(\omega) = \frac{1}{\sqrt{2\pi}} e^{-\omega^2/4a} \int_{-T-i\omega/2a}^{T-i\omega/2a} e^{-s^2} ds$$

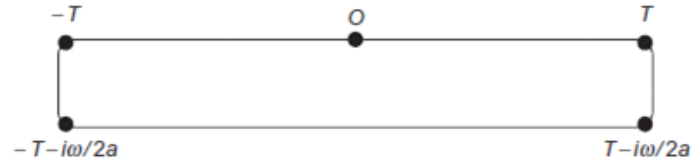


FIGURE 20.3 Contour for transform of Gaussian in Example 20.2.2.

$$g(\omega) = \frac{1}{\sqrt{2a}} e^{-\omega^2/4a^2}$$

Inverse Fourier Transform

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$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt$$

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{-i\omega t} d\omega$$

$$\begin{aligned} \phi_t(\omega) &= \frac{1}{\sqrt{2\pi}} e^{i\omega t} \\ \phi_t^*(\omega) &= \frac{1}{\sqrt{2\pi}} e^{-i\omega t} \end{aligned}$$

Fourier Integral

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{-i\omega x} \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt$$

$$f(x) = \int_{-\infty}^{\infty} dt f(t) \delta(x-t)$$

$$\delta(x-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(t-x)} d\omega$$

3D FT

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$$g(\vec{k}) = \frac{1}{(2\pi)^{3/2}} \int f(\vec{r}) e^{i\vec{k} \cdot \vec{r}} d^3 r$$

$$f(\vec{r}) = \frac{1}{(2\pi)^{3/2}} \int g(\vec{k}) e^{-i\vec{k} \cdot \vec{r}} d^3 k$$

Ex 20.2.5

$$\left[\frac{e^{-\alpha r}}{r} \right]^T(\vec{k}) = \frac{1}{(2\pi)^{3/2}} \frac{4\pi}{k^2 + \alpha^2}$$

$$\left[\frac{1}{r} \right]^T(\vec{k}) = \frac{1}{(2\pi)^{3/2}} \frac{4\pi}{k^2}$$

$$\left[\frac{1}{r^2} \right]^T(\vec{k}) = \frac{1}{(2\pi)^{3/2}} \frac{4\pi}{k^2}$$

$$\left[e^{-Zr} \right]^T(\vec{k}) = \frac{1}{(2\pi)^{3/2}} \frac{8\pi Z}{(k^2 + Z^2)^2}$$

$$\left[e^{-ar^4} \right]^T(\vec{k}) = \frac{1}{(2a)^{3/2}} e^{-k^2/4a}$$

Properties of FT

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$$g(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int f(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}} d^3r,$$

$$f(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int g(\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{r}} d^3k.$$

$$\left[f(\mathbf{r} - \mathbf{R}) \right]^T(\mathbf{k}) = e^{i\mathbf{k}\cdot\mathbf{R}} g(\mathbf{k}), \quad \text{(translation),} \quad (20.48)$$

$$\left[f(\alpha\mathbf{r}) \right]^T(\mathbf{k}) = \frac{1}{\alpha^3} g(\alpha^{-1}\mathbf{k}), \quad \text{(change of scale),} \quad (20.49)$$

$$\left[f(-\mathbf{r}) \right]^T(\mathbf{k}) = g(-\mathbf{k}), \quad \text{(sign change),} \quad (20.50)$$

$$\left[f^*(-\mathbf{r}) \right]^T(\mathbf{k}) = g^*(\mathbf{k}), \quad \text{(complex conjugation),} \quad (20.51)$$

$$\left[\nabla f(\mathbf{r}) \right]^T(\mathbf{k}) = -i\mathbf{k} g(\mathbf{k}), \quad \text{(gradient),} \quad (20.52)$$

$$\left[\nabla^2 f(\mathbf{r}) \right]^T(\mathbf{k}) = -k^2 g(\mathbf{k}), \quad \text{(Laplacian).} \quad (20.53)$$

20.4 Fourier Convolution Thm

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$f(x)$ & $g(x) \Rightarrow$ Convolution

$$(g * f)(x) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(y) g(x-y) dy$$

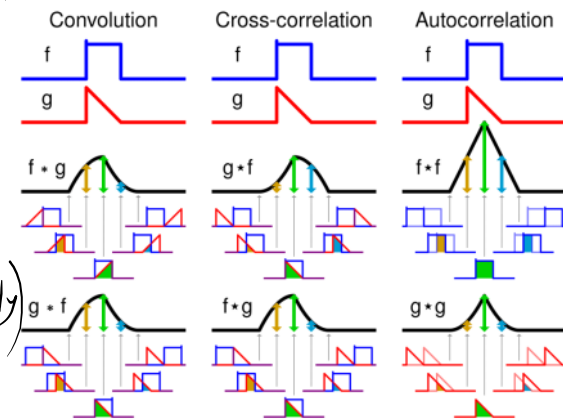
$$(g * f)(\vec{r}) = \frac{1}{(2\pi)^{3/2}} \int f(\vec{r}') g(\vec{r} - \vec{r}') d^3 r'$$

$$[(g * f)(x)]^T(t) = \frac{1}{\sqrt{2\pi}} \int e^{itx} dx \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(y) g(x-y) dy \right)$$

$$= \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{it(x-y)} g(x-y) dx \right) \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(y) e^{ity} dy \right)$$

$$= C_g(t) F(t)$$

$$e^{it(x-y)} = e^{itx} e^{-ity}$$



$$(f * g)(x) = \int_{-\infty}^{\infty} G(t) F(t) e^{-ixt} dt$$

$$\sum |a_n|^2 = \int |f|^2 dx$$

Parsaval's Relation

$x=0$ $\int_{-\infty}^{\infty} g(-y) f(y) dy = \int_{-\infty}^{\infty} G(t) F(t) dt$

Replace $g(-y) \Rightarrow g^*(y)$

$$\int_{-\infty}^{\infty} g^*(y) f(y) dy = \int_{-\infty}^{\infty} G^*(t) F(t) dt$$

$$\langle g | f \rangle = \langle Fg | Ff \rangle = \langle g | \underbrace{F^\dagger F}_{\text{unitary}} | f \rangle$$

20.7 Laplace Transform

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$$f(s) = \mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt$$

may diverge

$$\mathcal{L}\{F'(t)\} = s f(s) - F(0)$$

$$\mathcal{L}\{F''(t)\} = s^2 f(s) - s F(0) - F'(0)$$

$$\mathcal{L}\{F^{(n)}(t)\} = s^n f(s) - s^{n-1} F(0) - \dots + F^{(n-1)}(0)$$

Table 20.1 Laplace Transforms^a

$f(s)$	$F(t)$	Limitation	Equation
1.	$\delta(t)$	Singularity at $+0$	(20.140)
2.	$\frac{1}{s}$	$s > 0$	(20.129)
3.	$\frac{\Gamma(n+1)}{s^{n+1}}$	$s > 0, n > -1$	(20.136)
4.	$\frac{1}{s-k}$	$s > k$	(20.130)
5.	$\frac{1}{(s-k)^2}$	$s > k$	(20.176)
6.	$\frac{s}{s^2+k^2}$	$s > k$	(20.132)
7.	$\frac{k}{s^2+k^2}$	$s > k$	(20.133)
8.	$\frac{s}{s^2-k^2}$	$s > 0$	(20.134)
9.	$\frac{k}{s^2+k^2}$	$s > 0$	(20.135)
10.	$\frac{s-a}{(s-a)^2+k^2}$	$s > a$	(20.159)
11.	$\frac{k}{(s-a)^2+k^2}$	$s > a$	(20.158)
12.	$\frac{s^2-k^2}{(s^2+k^2)^2}$	$s > 0$	(20.177)
13.	$\frac{2ks}{(s^2+k^2)^2}$	$s > 0$	(20.178)
14.	$(s^2+a^2)^{-1/2}$	$s > 0$	(20.182)
15.	$(s^2-a^2)^{-1/2}$	$s > a$	Exercise 20.8.13
16.	$\frac{1}{a} \cot^{-1}\left(\frac{s}{a}\right)$	$s > 0$	Exercise 20.8.14
17.	$\left. \begin{aligned} \frac{1}{2a} \ln \frac{s+a}{s-a} \\ \frac{1}{a} \coth^{-1}\left(\frac{s}{a}\right) \end{aligned} \right\}$	$s > a$	Exercise 20.8.14
18.	$\frac{(s-a)^n}{s^{n+1}}$	$s > 0$	Exercise 20.8.16
19.	$\frac{1}{s} \ln(s+1)$	$s > 0$	Exercise 20.8.17
20.	$\frac{\ln s}{s}$	$-\ln t - \gamma$	Exercise 20.10.9

^a γ is the Euler-Mascheroni constant.

Ex 8H0

$$m \ddot{X}(t) + k X(t) = 0, \quad X(0) = X_0, \quad \dot{X}(0) = 0$$

$$m [s^2 X(s) - s X_0] + k X(s) = 0$$

$$X(s) = X_0 \frac{s}{s^2 + \omega_c^2}$$

$$X(t) = X_0 \cos(\omega_c t)$$

$$\omega_c^2 = \frac{k}{m}$$

Table 20.2 Laplace Transform Operations

	Operation	Equation
1. Laplace transform	$f(s) = \mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt$	(15.99)
2. Transform of derivative	$s f(s) - F(+0) = \mathcal{L}\{F'(t)\}$	(15.123)
	$s^2 f(s) - s F(+0) - F'(+0) = \mathcal{L}\{F''(t)\}$	(15.124)
3. Transform of integral	$\frac{1}{s} f(s) = \mathcal{L}\left\{\int_0^t F(x) dx\right\}$	Exercise 20.9.1
4. Change of scale	$\frac{1}{a} f\left(\frac{s}{a}\right) = \mathcal{L}\{F(at)\}$	(20.156)
5. Substitution	$f(s-a) = \mathcal{L}\{e^{at} F(t)\}$	(15.152)
6. Translation	$e^{-bs} f(s) = \mathcal{L}\{F(t-b)\}$	(15.164)
7. Derivative of transform	$f^{(n)}(s) = \mathcal{L}\{(-t)^n F(t)\}$	(15.173)
8. Integral of transform	$\int_s^{\infty} f(x) dx = \mathcal{L}\left\{\frac{F(t)}{t}\right\}$	(15.189)
9. Convolution	$f_1(s) f_2(s) = \mathcal{L}\left\{\int_0^t F_1(t-z) F_2(z) dz\right\}$	(15.193)
10. Inverse transform, Bromwich integral ²	$\frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} e^{st} f(s) ds = F(t)$	(15.212)

^a β must be large enough that $e^{-\beta t} F(t)$ vanishes as $t \rightarrow +\infty$.

20.10 Bromwich Integral

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$$F(t) = \mathcal{L}^{-1}\{f(s)\}$$

$$F(t) \text{ might diverge} \Rightarrow F(t) = e^{\beta t} G(t) \sim e^{\alpha t}$$

$$\beta > \alpha \Rightarrow G(t) \text{ convergent}$$

Fourier Integral of $G(t)$:

$$e^{\beta t} G(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iut} du \int_0^{\infty} \frac{G(v) e^{-iuv}}{e^{-\beta v} F(v)} dv$$

$$s = \beta + iu$$

$$F(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(iu + \beta)^t}{e} du \int_0^{\infty} F(v) e^{-(iu + \beta)v} dv$$

$$F(s) = \int_0^{\infty} e^{-sv} F(v) dv; \operatorname{Re}(s) > \beta$$

$$F(t) = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} e^{st} f(s) ds$$

Ex 20.201

$$f(s) = \frac{a}{s^2 - a^2} ; F(t) = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} \frac{a e^{st}}{s^2 - a^2} ds$$

$$\begin{aligned} F(t) &= \frac{1}{2\pi i} 2\pi i \left(\text{Res}_{s=a} + \text{Res}_{s=-a} \right) \\ &= \frac{a e^{at}}{2a} + \frac{a e^{-at}}{-2a} = \frac{1}{2} (e^{at} - e^{-at}) \\ &= \sinh(at) \end{aligned}$$

$$\underline{\text{Re}(s) \geq \beta}$$

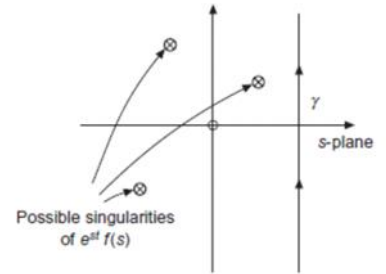


FIGURE 20.20 Possible singularities of $e^{st} f(s)$.

